HATCHSIDE · USER REQUIREMENTS

HatchSide user requirements

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-05	Amish Chadha	First user requirements, written at the kit 1.9.1 upgrade from the users and stakeholders in HTS-PRB-001 Tables 1 and 2; desk draft, not yet reviewed with a co-design partner

HatchSide user requirements

What the people who use and live around the kit need, written from their side. They come from the users and stakeholders in HTS-PRB-001 (Tables 1 and 2) and from the five tasks the kit exists for. They are a desk draft: no crew, utility or biogas programme has reviewed them yet, and the co-design partners in HTS-PRB-001 are candidates, none agreed. Each need traces to one or more product requirements in HTS-PRS-001 (matrix: HTS-RTM-001). Method: STANDARDS section 22.

Table 1. User requirements.

ID	Category	Title	Need	Priority	Source
UR-001	Safety	Nobody goes in	Routine clearing, sampling, raking and recovery are done from outside the space, so no job needs a person to enter	High	Sanitation workers, municipal crews (PRB Table 1)
UR-002	Performance	Fits the openings we meet	The kit sets up over, and every tool passes through, the common street manhole, tank hatch and digester opening	High	Sanitation workers, biogas technicians (PRB Table 1)
UR-003	Performance	Works at our depths	Tools reach the working depth of sewers, tanks and household digesters, and the crew knows how deep the tool is	High	Municipal crews, biogas technicians
UR-004	Performance	Does the five jobs	One kit clears silt, takes samples, rakes, clips a line to a casualty and reseals a digester dome	High	All user types (PRB Table 1)
UR-005	Usability	Up in minutes	A crew of two or three sets the kit up before the job in a few minutes, without special tools	Medium	Municipal crews
UR-006	Logistics	Carried by the crew	The kit travels in loads one person can carry, on foot, by handcart or on a motorbike	High	Sanitation workers, biogas technicians
UR-007	Cost	Affordable	A municipality or a farmer cooperative can afford one, well under the price of a commercial tripod and winch	High	All user types, funders
UR-008	Safety	Never a lifeline, never tips	The kit cannot be mistaken for equipment that lifts people, holds its load without failing, and does not tip if the rope pulls sideways	High	Sanitation workers, regulators, safety teams
UR-009	Durability	Lasts in sewage and sun	Parts survive sewage splash, hydrogen sulfide air, heat and monsoon rain	Medium	Municipal crews, industrial contractors
UR-010	Usability	Changes heads fast	A crew changes from one tool head to another in about a minute without tools	Medium	Municipal crews
UR-011	Maintenance	Made and mended locally	A local workshop with a welder and drill can make and mend every part from stock sizes	Medium	Local fabricators, municipal workshops
UR-012	Safety	Guards the opening	The open manhole is guarded against falls in and against traffic while the kit is set up	Medium	Municipal crews, public at the opening
UR-013	Usability	Easy for one person to work	One person can raise and lower a full tool from the winch without strain	Medium	Sanitation workers
UR-014	Performance	Samples tell us where	A sample can be tied to the depth it came from, to within about the length of the sampler	Low	Municipal crews, industrial contractors

Priority: High (weight 3 in concept scoring), Medium (2), Low (1).

Notes

- UR-008 is a safety need first: the winch lifts material and tool heads only. Nobody is lowered or raised on it, and the retrieval head only clips a line to a casualty, with the haul made on certified rescue equipment (HTS-DDR-001, D2).
- UR-007 is read against the USD 5,000 value-engineering target in `project.yaml`, a hypothetical control target and not a spending limit (STANDARDS section 18).
- UR-009 cannot be closed on paper: corrosion life is a TRL 4 test.